Geometric Origin of CP Phases from Hyperbolic Holonomy

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Abstract

We show that CP-violating phases arise naturally as geometric holonomy invariants associated with closed geodesics in compact hyperbolic 3-manifolds. For a discrete group acting properly discontinuously and cocompactly on $\mathbb{H}^3$, an orientation-reversing isometry induces complex conjugation on all holonomy phases of a fixed background U(1) connection. Any geodesic whose holonomy is not ± 1 therefore witnesses CP violation that is rephasing-invariant, topologically stable, and independent of continuous parameter tuning. We make the argument precise, prove a clean stability theorem for flat connections, and illustrate the mechanism with the Weeks manifold.

1 Introduction

CP violation in the Standard Model is parametrized by complex phases in Yukawa couplings or mixing matrices. Rephasing-invariant combinations such as the Jarlskog invariant [3] isolate the physical content, but their origin remains unexplained by the Standard Model itself. In this paper we show that CP-violating phases can arise from geometry alone, without assumptions about flavor dynamics or continuous parameter choices.

The mechanism is simple. In a compact hyperbolic 3-manifold M every closed geodesic carries a holonomy phase of a background U(1) connection. If M admits an orientation-reversing isometry Θ, then Θ acts on each holonomy by complex conjugation: the phase flips sign. A geodesic with holonomy $W(\gamma) \notin \{\pm 1\}$ therefore provides an observable that distinguishes geometry from its mirror image.

The paper is organized as follows. Section 2 sets up hyperbolic holonomy. Section 3 proves the orientation-reversal lemma and the main CP theorem. Section 4 establishes rephasing invariance. Section 5 proves topological stability for flat connections. Section 6 discusses the Weeks manifold example. Section 7 discusses future directions.

2 Hyperbolic Holonomy

Let Γ be a torsion-free discrete group acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries, so that $M = \mathbb{H}^3/\Gamma$ is a compact oriented hyperbolic 3-manifold. Fix a basepoint $x_0 \in M$.

Definition 2.1 (Closed geodesic)**.** A *closed geodesic* in M corresponds to a nontrivial conjugacy class $[\gamma] \subset \Gamma$ where γ is a hyperbolic element. Its length is $\ell(\gamma) = d(x, \gamma \cdot x)$ for any x on the axis.

Let A be a fixed smooth U(1) connection on M. Its holonomy around a loop c based at x_0 is

$$W(c) = \exp\Big(i \oint_c A\Big) \in \mathrm{U}(1). \tag{1}$$

Since U(1) is abelian, $W(c)$ depends only on the free homotopy class of c; we write $W(\gamma)$ for this invariant.

3 Orientation Reversal and CP

Let $\Theta \colon M \to M$ be an orientation-reversing isometry.

Lemma 3.1 (Holonomy conjugation under orientation reversal)**.** *Let A be a fixed background* U(1) *connection on M, and let $\Theta \colon M \to M$ be an orientation-reversing isometry. For every hyperbolic $\gamma \in \Gamma$,*

$$W\big(\Theta(\gamma)\big) = W(\gamma)^{-1} = \overline{W(\gamma)}. \tag{2}$$

Proof. **Step 1 (geometric).** Since Θ is an isometry, it maps the closed geodesic c_γ to the closed geodesic $c_{\Theta(\gamma)}$ of the same length.

Step 2 (orientation). Since Θ reverses orientation, it reverses the orientation of every curve. Therefore

$$\oint_{\Theta(c_\gamma)} A = \oint_{c_\gamma} \Theta^* A = -\oint_{c_\gamma} A, \tag{3}$$

where the second equality uses $\Theta^* A = -A$ for an orientation-reversing isometry acting on a U(1) connection. Exponentiating gives $W(\Theta(\gamma)) = W(\gamma)^{-1} = \overline{W(\gamma)}$. □

Theorem 3.2 (Geometric CP violation)**.** *Fix a background* U(1) *connection A on M. If M admits an orientation-reversing isometry Θ and there exists a hyperbolic $\gamma \in \Gamma$ with $W(\gamma) \notin \{\pm 1\}$, then $\arg W(\gamma) \neq 0, \pi$ is a rephasing-invariant observable that changes sign under Θ.*

Proof. We work with a fixed background connection A; we do not optimize over the space of connections. The Wilson loop $W(\gamma)$ is invariant under all local gauge transformations $A \to A + d\lambda$ because $\oint d\lambda = \lambda(x_0) - \lambda(x_0) = 0$. By Lemma 3.1, $\arg W(\Theta(\gamma)) = -\arg W(\gamma)$. If $W(\gamma) \notin \{\pm 1\}$ the phase is nonzero and no field redefinition can simultaneously set it to zero and respect the geometry. □

4 Rephasing Invariance

Proposition 4.1 (Rephasing invariance)**.** *Holonomy phases $W(\gamma)$ are invariant under all smooth* U(1) *gauge transformations $A \mapsto A + d\lambda$.*

Proof. $W(\gamma) \mapsto W(\gamma) \cdot \exp(i \oint_{c_\gamma} d\lambda) = W(\gamma) \cdot e^{i(\lambda(x_0) - \lambda(x_0))} = W(\gamma)$. □

5 Topological Stability for Flat Connections

Theorem 5.1 (Topological stability)**.** *Let A be a flat* $\mathrm{U}(1)$ *connection on M, classified by a representation $\rho\colon \pi_1(M) \to \mathrm{U}(1)$. For any hyperbolic $\gamma \in \Gamma$, $W(\gamma) = \rho([\gamma])$, which:*

(i) *is invariant under all metric deformations preserving topological type;*

(ii) *is invariant under continuous deformations of A within its gauge equivalence class;*

(iii) *gives a stable CP-odd observable under any deformation satisfying* (i), (ii)*, and preserving Θ.*

Proof. For a flat connection, parallel transport depends only on homotopy class, so $W(\gamma) = \rho([\gamma])$ is purely topological. Statement (i): metric deformations do not change $\pi_1(M)$ or ρ. Statement (ii): within a gauge class, ρ is constant. Statement (iii) follows from Theorem 3.2. □

6 Example: The Weeks Manifold

The Weeks manifold [4] is the compact hyperbolic 3-manifold of smallest known volume ($v \approx 0.9427$). It admits an orientation-reversing involution [2], and its first homology is $H_1(M_W; \mathbb{Z}) \cong \mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/5$ [1], admitting nontrivial characters $\chi\colon H_1 \to \mathrm{U}(1)$ sending generators to fifth roots of unity. Setting $W(\gamma) = e^{2\pi i/5}$ for an appropriate generator gives $\arg W(\gamma) = 2\pi/5 \neq 0, \pi$, realizing a geometric CP-odd observable.

7 Relation to Physical CP Observables: Outlook

The geometric mechanism described here produces rephasing-invariant, topologically stable CP-odd phases. In a concrete physical model such phases could arise from a hidden-sector U(1) gauge field whose holonomy around compact extra dimensions enters effective Yukawa couplings through boundary conditions. The holonomy phases would then contribute to rephasing invariants such as the Jarlskog determinant [3]. Constructing a realistic embedding is a substantial further project left for future investigation.

8 Conclusion

We have shown that CP violation can arise from geometry: holonomy phases of a background U(1) connection on a compact hyperbolic 3-manifold, combined with an orientation-reversing isometry, produce rephasing-invariant CP-odd observables. For flat connections these phases are topological invariants stable under all metric deformations. The Weeks manifold provides a concrete example realizing the mechanism at fifth roots of unity. The construction requires no continuous parameter tuning and applies to any compact hyperbolic 3-manifold admitting both a nontrivial flat U(1) connection and an orientation-reversing isometry.

References

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